

mdstats 0.20.215a0 - TRAIN2 FP32 CuEq parity hotfix

Scope

This release changes only the TRAIN2 FP32 e3nn-versus-pure-CuEq numerical-equivalence policy and the release provenance that binds it. It does not change the source/DATA6 parity policy, FP64 parity, training convergence criteria, replay logic, target-data selection, or FINAL-GPU1 matrix membership.

Policy change

- Generic source/DATA6 FP32 parity: unchanged at $\text{rtol}=1\text{e-}5$, $\text{atol}=1\text{e-}6$.
- TRAIN2 FP32 parity: $\text{rtol}=1\text{e-}5$, $\text{atol}=1\text{e-}5$ (previously $\text{atol}=2\text{e-}6$).
- TRAIN2/source FP64 parity: unchanged at $\text{rtol}=1\text{e-}10$, $\text{atol}=1\text{e-}12$.

The revised TRAIN2 ceiling is motivated by MPA-0/default workstation evidence ($E_{\text{max}}=2.384\text{e-}7$, $F_{\text{max}}=8.911\text{e-}6$, $S_{\text{max}}=1.660\text{e-}7$, $D_{\text{max}}=2.883\text{e-}7$, $\text{selection_identical}=\text{True}$). It is treated as an FP32 backend-equivalence envelope for non-associative/reordered accumulation, not as an accuracy or convergence tolerance.

Fail-closed behavior

- Selection identity is still mandatory.
- Non-finite values still fail immediately.
- The $1\text{e-}5$ ceiling is fixed and never automatically widened.
- A zero-reference difference just above $1\text{e-}5$ is explicitly regression-tested as a failure.
- No automatic e3nn fallback is introduced.

FINAL-GPU1 binding

FINAL-GPU1 remains v3/18 items. Preflight advances from v8 to v9 and now embeds the exact TRAIN2 parity-policy payload and digest in the handoff manifest. Handoff integrity verification rejects a policy-digest mismatch.